LearnAble Study Notes

Cybersecurity

Note: No external sources were supplied for this topic. Therefore, these study notes are labeled as general educational content, not research-verified content.

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Cybersecurity: Student Study Notes

1. Topic Overview

Cybersecurity is the practice of protecting systems, networks, programs, devices, and data from digital attacks, unauthorized access, damage, or theft. In our increasingly digital world, cybersecurity is essential for safeguarding personal privacy, financial information, national security, and critical infrastructure (such as power grids and healthcare systems).

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2. Key Concepts and Definitions

* **Confidentiality:** Ensuring that sensitive information is accessed only by authorized individuals and kept away from those not permitted to see it.

* **Integrity:** Ensuring that data and systems are accurate, complete, and trustworthy, meaning they have not been tampered with or altered by unauthorized parties.

* **Availability:** Ensuring that information, network resources, and services are accessible to authorized users whenever they are needed.

* **Malware (Malicious Software):** Software specifically designed to disrupt, damage, or gain unauthorized access to a computer system (e.g., viruses, ransomware, spyware).

* **Phishing:** A social engineering attack where criminals trick individuals into revealing sensitive information—such as passwords or credit card numbers—by masquerading as a trustworthy entity.

* **Authentication:** The process of verifying the identity of a user, device, or system before allowing access to resources.

* **Encryption:** The process of converting readable data into an unreadable format (ciphertext) using an algorithm and a key, protecting it from being read if intercepted.

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3. Detailed Explanations

The CIA Triad

The foundational framework of cybersecurity is known as the **CIA Triad** (Confidentiality, Integrity, Availability).

* **Confidentiality** relies heavily on tools like encryption and access controls (passwords, biometrics) so that data remains secret.

* **Integrity** relies on checksums, version controls, and digital signatures to verify that data has not been altered maliciously or accidentally during transit or storage.

* **Availability** relies on hardware maintenance, system backups, and network resistance to attacks (like Distributed Denial of Service, or DDoS) to ensure systems stay online.

Threat Vectors and Human Error

Many cyber attacks do not exploit technical flaws in software; instead, they exploit human behavior. This is known as **social engineering**. Attackers manipulate human psychology—using fear, urgency, or curiosity—to trick people into handing over credentials or clicking malicious links. Cybersecurity professionals often state that "users are the weakest link," which is why security awareness training is just as important as installing firewalls and antivirus software.

Defense-in-Depth

Rather than relying on a single security measure, cybersecurity utilizes a strategy called **Defense-in-Depth**. This means layering multiple security controls so that if one fails, others are still in place to stop an attacker. Examples include combining firewalls, multi-factor authentication (MFA), regular software updates, and employee training.

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4. Examples

* **Ransomware Attack:** A hospital's computer system is infected with malware that encrypts all patient files. The attackers demand a financial payment (ransom) in exchange for the decryption key, disrupting medical operations and endangering patient care.

* **Phishing Email:** An employee receives an email that looks exactly like a message from their company’s IT department, stating their password is about to expire and they must click a link to reset it. The link leads to a fake login page designed to steal their credentials.

* **Multi-Factor Authentication (MFA):** When logging into a student portal, you enter your password (something you know) and then approve a notification sent to your smartphone (something you have). Even if a hacker steals your password, they cannot log in without your phone.

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5. Summary

* Cybersecurity protects digital systems, networks, and data from attacks and theft.

* The **CIA Triad** (Confidentiality, Integrity, Availability) forms the core goals of security measures.

* Attacks often target humans through **social engineering** (like phishing) rather than just targeting code.

* **Defense-in-Depth** uses multiple layers of security (such as encryption, MFA, and backups) to keep systems safe.

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6. Five Practice Questions with Answers

1. What does the "A" stand for in the CIA Triad, and what does it mean?

* **Answer:** The "A" stands for **Availability**. It means that information, network resources, and services must be accessible to authorized users whenever they need them.

2. Explain how a phishing attack works.

* **Answer:** A phishing attack is a form of social engineering where an attacker masquerades as a trustworthy entity (like a bank, school, or IT department) via email, text, or message to trick a user into revealing sensitive information, such as passwords or financial data, or clicking a malicious link.

3. Why is Multi-Factor Authentication (MFA) more secure than using a password alone?

* **Answer:** MFA requires two or more verification methods from different categories (e.g., something you know like a password, plus something you have like a code sent to your phone). If a hacker manages to steal just your password, they still cannot access the account without the second factor.

4. What is the primary purpose of data encryption?

* **Answer:** The primary purpose of encryption is to convert readable data into an unreadable format using a mathematical algorithm and a key. This ensures that even if unauthorized parties intercept the data, they cannot read or use it.

5. Why are humans often considered the "weakest link" in cybersecurity?

* **Answer:** Humans are often targeted by social engineering tactics (like manipulation, fear, or urgency) that exploit psychological traits rather than technical security flaws, leading people to willingly give up credentials or bypass security rules.